

OS COURSE PROJECT REPORT

TEAM NO: 9

Statement: Smart Parking Alerts with Multi-Zone Alerts

NAME	DIV	USN	ROLL NO
Soubhagya Hunderkar	E	01FE23BEC272	538
Shivani Solapur	E	01FE23BEC274	540
Srushti Rajput	E	01FE23BEC275	541
Khushi Tavakari	E	01FE23BEC280	546

1.Problem Statement:

The aim is to design a Smart Parking Assist system using the ARM Cortex-M3 based LPC1768 microcontroller. The system should continuously monitor the distance between the vehicle and nearby obstacles using an ultrasonic sensor. Based on the detected distance, the system must provide multi-level alerts to the driver using LED indicators. Each detection event must be logged and sent over UART for monitoring/debugging. If the obstacle is too close, the motor movement should be immediately stopped to prevent collision.

2.Introduction:

Parking safely in congested areas requires fast, reliable proximity sensing. A Smart Parking Assist system automates distance monitoring to help the driver avoid collisions. In this project, an ultrasonic sensor is interfaced with the LPC1768 microcontroller to measure object distance in real-time. Based on predefined threshold zones, the system alerts the driver using visual (LEDs) warnings. The design integrates distance measurement, RTOS-based task management, motor interlocking, and UART-based event logging. This makes the system efficient, scalable, and suitable for real automotive applications like parking automation, reverse assistance, and autonomous navigation support.

3.Problem Analysis:

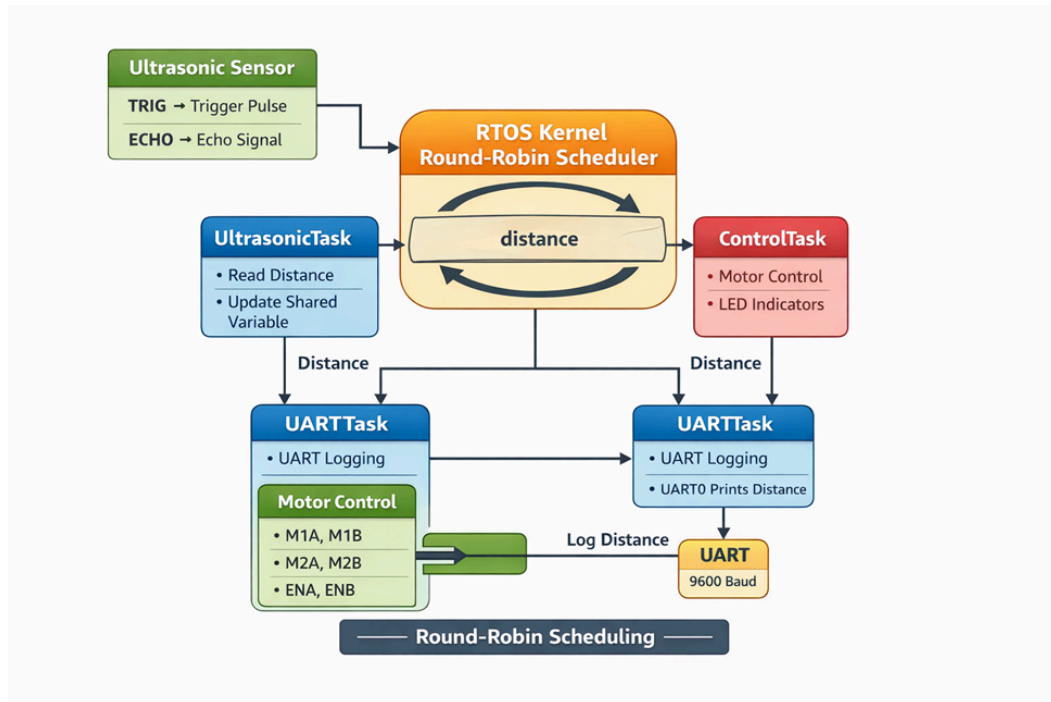
Features selected:

Feature	Description
Real-time distance sensing	Ultrasonic sensor continuously detects obstacle distance
Multi-Zone Alert System	LEDs indicate SAFE, CAUTION and DANGER zones
Motor Interlock	Motor stops automatically if the object is too near
UART Logging	Each alert/detection log transmitter to monitoring system
RTOS-based design	Scheduling Algorithm(Task Management)

4.Peripherals chosen for prototyping:

Peripheral	Purpose
Ultrasonic Sensor (Trigger/Echo Pins)	Distance Measurement
LEDs (Green/Yellow/Red)	Visual alert indication for zones
Motor Driver (L298)	Drives motor; stops upon near obstacle
UART Interface	Logs sensor readings and alert status
Timer (Capture Mode)	Precise calculation of echo pulse width
RTOS Kernel Objects	Synchronization and task scheduling

5. Block diagram:



6. Specifications of all Peripherals Used and Calculations:

Ultrasonic Sensor: HC-SR04

LED: Green, Red and White

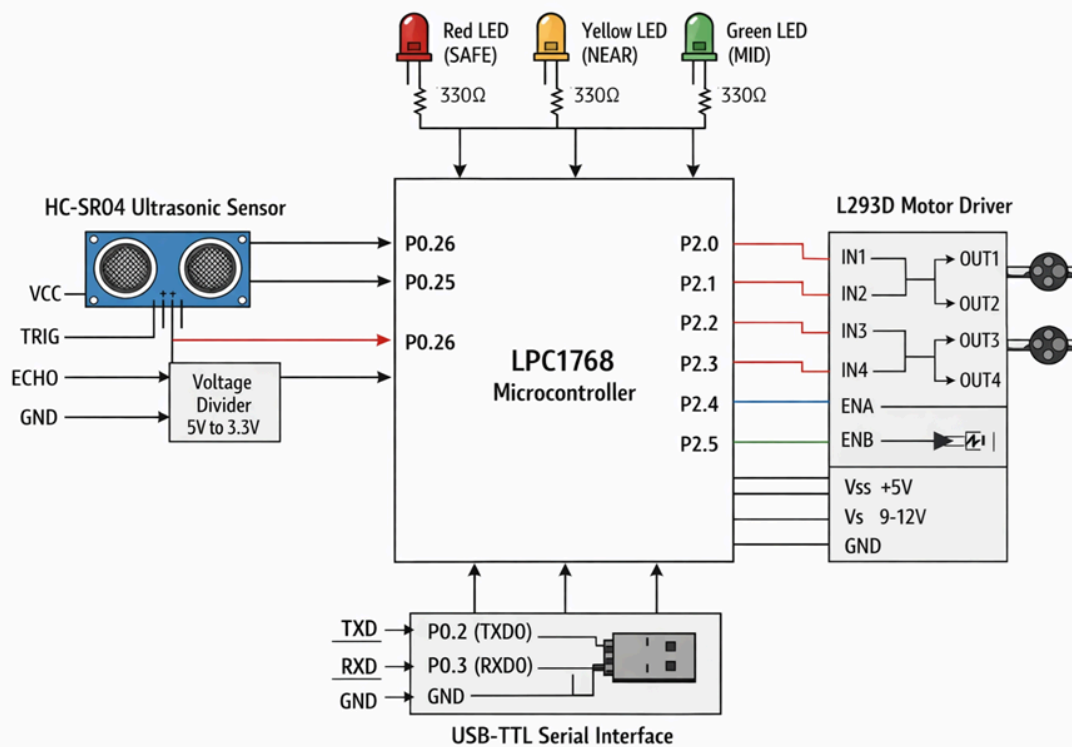
DC Motor: 12V

Motor Driver: L298

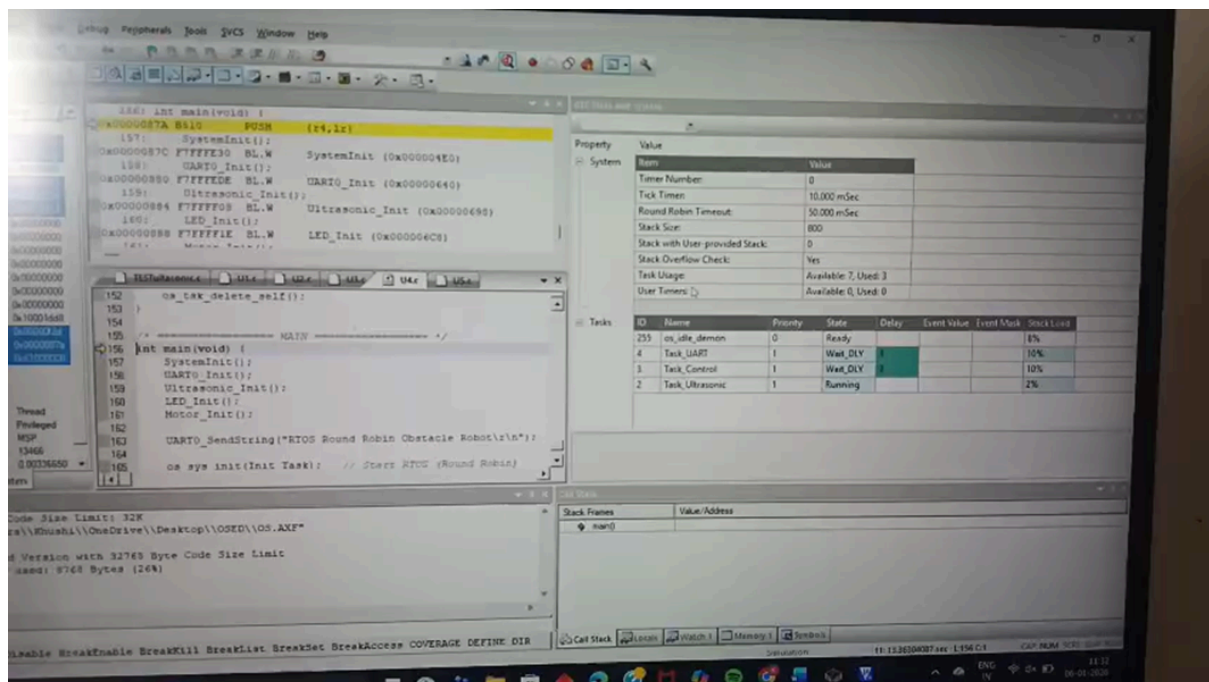
Microcontroller: LPC1768

Communication Parameter : 7200 or 9600 baud rate

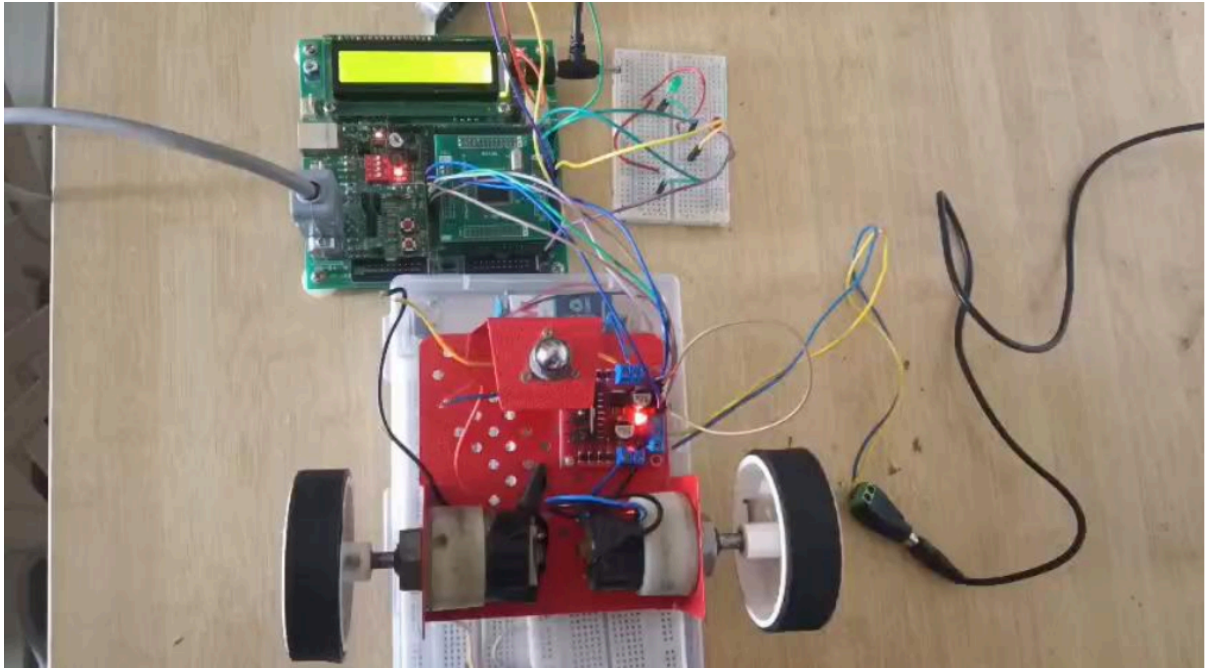
7. Interfacing Diagram:



8. Results:



9.Prototype:



Code:

```
Target 1
Source Group 1
  startup_LPC17xx.s
  system_LPC17xx.c
  RTX_Conf_CM.c
  U4.c

001 #include "LPC17xx.h"
002 #include <stdio.h>
003 #include <RTL.h> // RTX RTOS
004
005 #define TRIG (1 << 26)
006 #define ECHO (1 << 25)
007
008 #define LED_SAFE (1 << 18)
009 #define LED_NEAR (1 << 19)
010 #define LED_MID (1 << 20)
011
012 #define M1A (1 << 0)
013 #define M1B (1 << 1)
014 #define M2A (1 << 2)
015 #define M2B (1 << 3)
016 #define ENA (1 << 4)
017 #define ENB (1 << 5)
018
019 #define THRESHOLD_SAFE 10.0
020 #define THRESHOLD_NEAR 20.0
021 #define THRESHOLD_MID 30.0
022
023 /* ----- Shared Data (RTOS style) ----- */
024 float distance;
025 char uart_buf[50];
026
027 /* ----- Delay ----- */

Build Output
Build target 'Target 1'
compiling U4.c...
linking...
Program Size: Code=8456 RO-data=268 RW-data=44 ZI-data=7596
FromELF: creating hex file...
"OS.axf" - 0 Error(s), 0 Warning(s).
```

```

023  /* ----- Shared Data (RTOS style) ----- */
024  float distance;
025  char uart_buf[50];
026
027  /* ----- Delay ----- */
028  void delay_ms(unsigned int ms) {
029      unsigned int i, j;
030      for(i=0;i<ms;i++)
031          for(j=0;j<600;j++);
032  }
033
034  /* ----- UART ----- */
035  void UART0_Init(void) {
036      LPC_SC->PCONP |= (1 << 3);
037      LPC_PINCON->PINSEL0 &= ~( (3<<4) | (3<<6) );
038      LPC_PINCON->PINSEL0 |= ( (1<<4) | (1<<6) );
039      LPC_UART0->LCR = 0x83;
040      LPC_UART0->DLL = 162;
041      LPC_UART0->DLM = 0;
042      LPC_UART0->LCR = 0x03;
043  }
044
045  void UART0_SendString(char *str) {
046      while(*str) {
047          while(!(LPC_UART0->LSR & (1<<5)));
048          LPC_UART0->THR = *str++;
049      }
050  }
051

```

```

052  /* ----- Init ----- */
053  void Ultrasonic_Init(void) {
054      LPC_GPIO0->FIODIR |= TRIG;
055      LPC_GPIO0->FIODIR &= ~ECHO;
056      LPC_SC->PCONP |= (1 << 1);
057      LPC_TIM0->PR = 24;
058  }
059
060  void LED_Init(void) {
061      LPC_GPIO1->FIODIR |= LED_SAFE | LED_NEAR | LED_MID;
062      LPC_GPIO1->FIOCLR = LED_SAFE | LED_NEAR | LED_MID;
063  }
064
065  void Motor_Init(void) {
066      LPC_GPIO2->FIODIR |= M1A | M1B | M2A | M2B | ENA | ENB;
067      LPC_GPIO2->FIOCLR = ENA | ENB;
068  }
069
070  /* ----- Motor ----- */
071  void Motor_Forward(void) {
072      LPC_GPIO2->FIOSET = M1A | M2A | ENA | ENB;
073      LPC_GPIO2->FIOCLR = M1B | M2B;
074  }
075
076  void Motor_Stop(void) {
077      LPC_GPIO2->FIOCLR = ENA | ENB;
078      LPC_GPIO2->FIOCLR = M1A | M1B | M2A | M2B;
079  }
080

```

```
080
081 /* ----- Ultrasonic ----- */
082 float Measure_Distance(void) {
083     unsigned int time;
084
085     LPC_GPIO0->FIOCLR = TRIG;
086     delay_ms(2);
087     LPC_GPIO0->FIOSET = TRIG;
088     delay_ms(1);
089     LPC_GPIO0->FIOCLR = TRIG;
090
091     while(!(LPC_GPIO0->FIOPIN & ECHO));
092     LPC_TIM0->TCR = 0x02;
093     LPC_TIM0->TCR = 0x01;
094
095     while(LPC_GPIO0->FIOPIN & ECHO);
096     LPC_TIM0->TCR = 0x00;
097
098     time = LPC_TIM0->TC;
099     return (time * 0.0343f) / 2.0f;
100 }
101
```

```

106  /* Task 1: Ultrasonic sensing */
107  task void Task_Ultrasonic(void) {
108      while(1) {
109          distance = Measure_Distance();
110          os_dly_wait(5); // Round Robin slice
111      }
112  }
113
114  /* Task 2: Motor + LED control */
115  task void Task_Control(void) {
116      while(1) {
117          LPC_GPIO1->FIOCLR = LED_SAFE | LED_NEAR | LED_MID;
118
119          if(distance <= THRESHOLD_SAFE) {
120              Motor_Stop();
121              LPC_GPIO1->FIOSET = LED_SAFE;
122          }
123          else if(distance <= THRESHOLD_NEAR) {
124              Motor_Stop();
125              LPC_GPIO1->FIOSET = LED_NEAR;
126          }
127          else if(distance <= THRESHOLD_MID) {
128              Motor_Forward();
129              LPC_GPIO1->FIOSET = LED_MID;
130          }
131          else {
132              Motor_Forward();
133          }
134          os_dly_wait(5);
135      }
136  }
137

```

```

137
138 /* Task 3: UART logging */
139 task void Task_UART(void) {
140     while(1) {
141         sprintf(uart_buf, "Distance = %.1f cm\r\n", distance);
142         UART0_SendString(uart_buf);
143         os_dly_wait(10);
144     }
145 }
146
147 /* ----- INIT TASK ----- */
148 task void Init_Task(void) {
149     os_tsk_create(Task_Ultrasonic, 1);
150     os_tsk_create(Task_Control, 1);
151     os_tsk_create(Task_UART, 1);
152     os_tsk_delete_self();
153 }
154
155 /* ===== MAIN ===== */
156 int main(void) {
157     SystemInit();
158     UART0_Init();
159     Ultrasonic_Init();
160     LED_Init();
161     Motor_Init();
162
163     UART0_SendString("RTOS Round Robin Obstacle Robot\r\n");
164
165     os_sys_init(Init_Task); // Start RTOS (Round Robin)
166     while(1);
167 }
168

```

10.Conclusion:

The Smart Parking Assist system implemented on the LPC1768 microcontroller successfully demonstrates an embedded solution for vehicle proximity detection using an ultrasonic sensor. By integrating multi-zone LED alerts and DC motor interlock, the system provides real-time feedback to prevent collisions, enhancing parking safety. The use of an RTOS with Round-Robin scheduling ensures efficient multitasking, allowing simultaneous execution of ultrasonic sensing, motor and LED control, and optional UART logging without task interference. The project highlights the effective use of shared variables and task prioritization to manage sensor data and actuator responses reliably. Overall, the system is robust, modular, and can be further extended to incorporate additional safety features, such as automated braking or multi-sensor arrays, demonstrating a practical approach to intelligent parking assistance.